MULTI-CLOUD FDE STACK · ISSUE 06 OF 08

Managed Postgres for RAG.

RDS, Cloud SQL, Azure Database for PostgreSQL, plus pgvector.

06

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Swipe →

Managed PostgreSQL

	HA	PASSWORDLESS AUTH
AWS RDS	Multi-AZ	RDS IAM auth
GCP Cloud SQL	Regional HA	IAM DB auth + Auth Proxy
Azure DB for PG	Zone-redundant HA	Entra ID auth

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pgvector in practice

01 Check the version

HNSW indexes need pgvector 0.5.0 or later.

02 Azure first step

Allowlist vector in azure.extensions.

03 Hybrid search

Combine full-text and vector scores, then re-rank.

04 Filter by tenant

Permissions in the WHERE clause, not after.

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When pgvector is enough

STAY ON POSTGRES

- Up to millions of chunks
- Need joins and filters
- Customer runs Postgres
- One fewer security review

LOOK FURTHER

- Hundreds of millions of vectors
- Strict latency at high QPS
- Heavy multi-tenant search
- Search team already exists

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INTERVIEW SCENARIO

Scaled to 40 pods. Database refused connections.

Every pod brought its own pool. Add a pooler, shrink pools and cap autoscaling.

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Database go-live

- [x] Private networking only
- [x] Passwordless auth per service
- [x] Pooler in front of the database
- [x] Backups and restore tested
- [x] Index and recall measured

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KEY TAKEAWAY

Use the database they already trust before adding a new one.

→ pgvector covers most pilots

→ Filter by permission in the query

→ Pool connections before you scale

YOUR TURN

Postgres with pgvector or a dedicated vector database for your last project?

FOLLOW AKHILESH RANJAN SINGH

Next in the series: #07 See it before the customer does

Save · Repost

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